

PAPER A

Science

2019

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

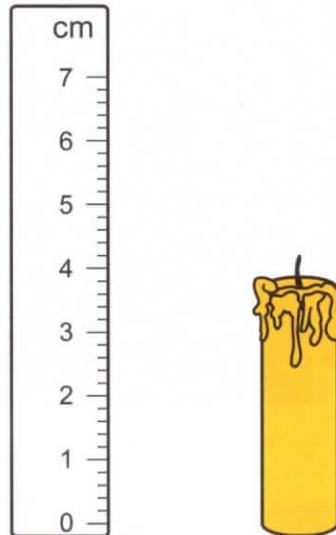
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Which of these animals has four legs?

- (A) a chicken (B) a mouse (C) a dolphin (D) a worm

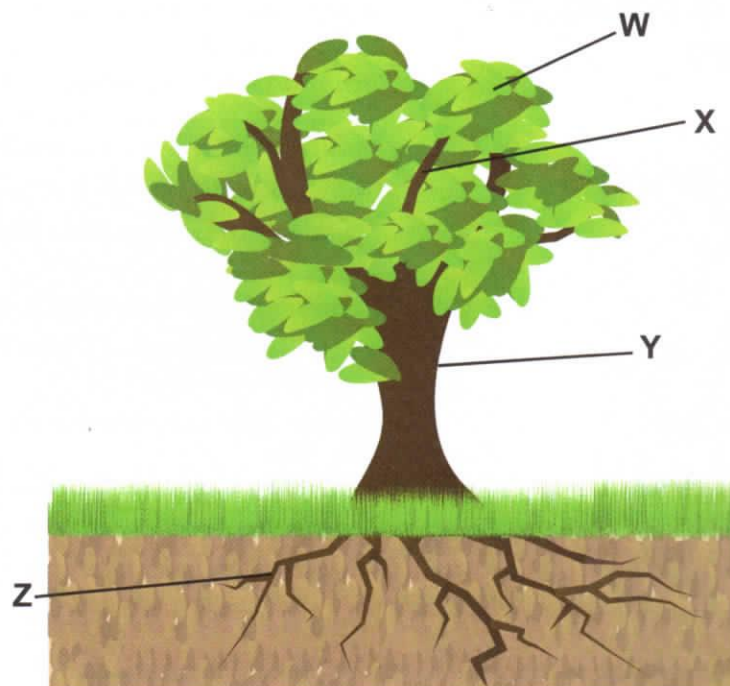
2. Milan measures the height of this candle with a ruler.



What is the height of the candle?

- (A) 7 cm (B) 6 cm (C) 4 cm (D) 1 cm

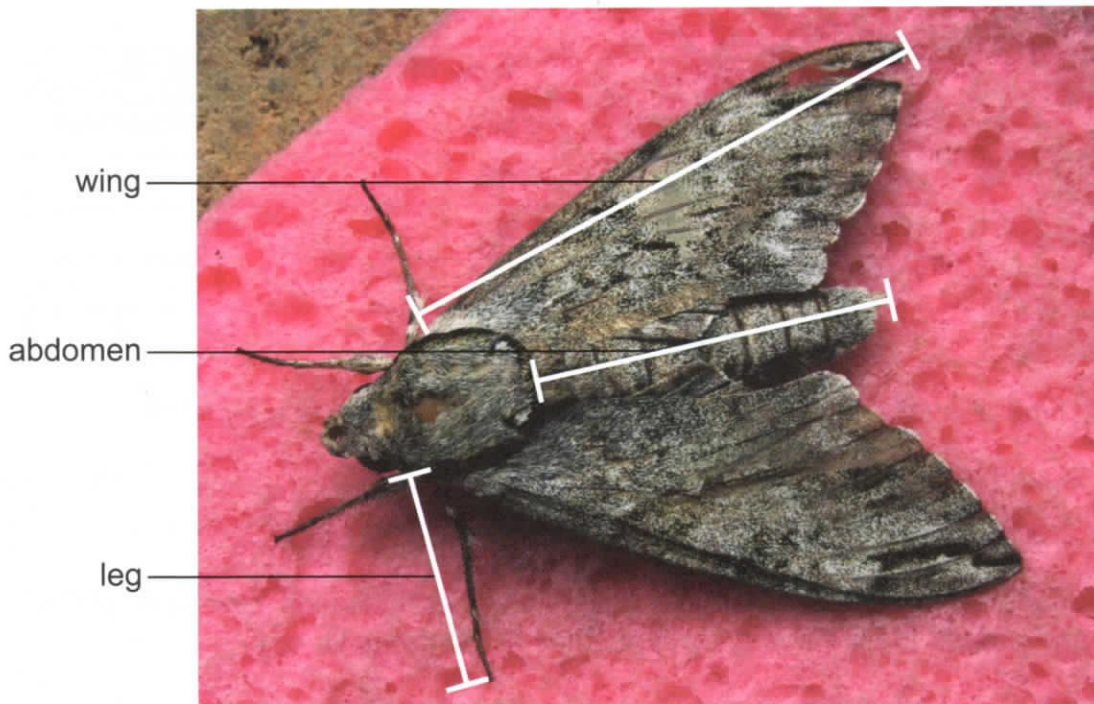
3. Trees need water from the ground to survive and grow.



Which part of a tree takes in water from soil?

- (A) W (B) X (C) Y (D) Z

4. This moth landed on a sponge. Didi used a ruler to measure the length of some parts of its body.



Which of the following statements is correct?

- (A) The leg, wing and abdomen are all the same length.
- (B) The abdomen is longer than the wing which is longer than the leg.
- (C) The leg is shorter than the wing which is longer than the abdomen.
- (D) The abdomen is shorter than the leg which is longer than the wing.

5. Mika filled a paper drinking cup with water.

Which of these weighs more than the cup of water?

- | | |
|-----------------|----------------------------------|
| (A) an ice cube | (B) a loaf of bread |
| (C) a pencil | (D) a page from an exercise book |

6. There are boys and girls at Diana's school. She wants to know what kind of pet the children in her school like most.

She does not have time to ask all the children at her school.

Which of these would be the best way for Diana to find out?

- (A) Ask her friends to name their favourite kind of pet.
- (B) Ask the girls in her class to name their favourite kind of pet.
- (C) Ask the boys in her year group to name their favourite kind of pet.
- (D) Ask some boys and girls from each year group to name their favourite kind of pet.

7. Dave looked at four machines in his kitchen.

Each machine changed electrical energy into other types of energy.

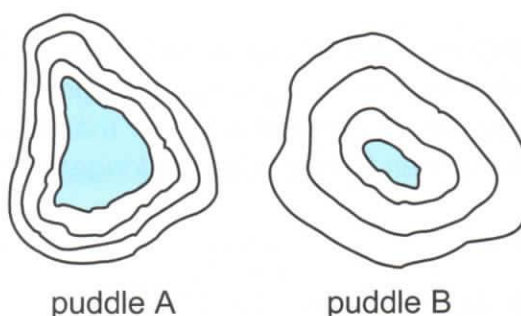
		Type of energy			
		Sound	Light	Heat	Movement
Type of Machine	Fridge	✓	✓	✓	
	Grill		✓	✓	
	Oven		✓	✓	
	Blender	✓		✓	✓

Which type of energy do all the machines change electrical energy into?

- (A) sound (B) light (C) heat (D) movement

8. Sarah made two puddles on her floor using two different liquids. She used 100 mL of each liquid and poured them into separate puddles at the same time.

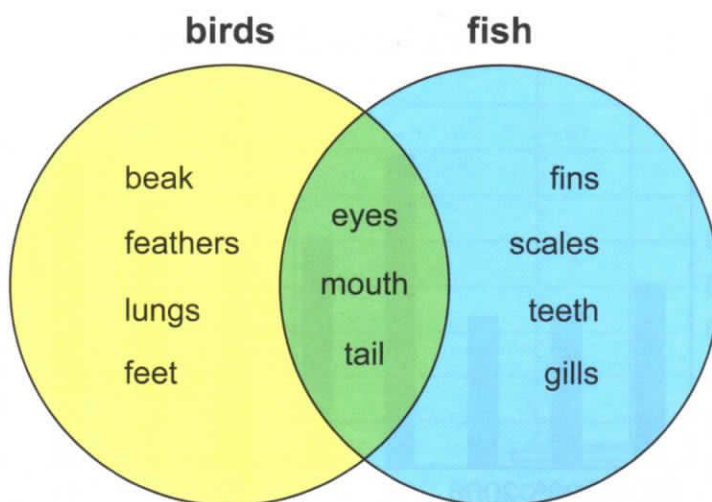
Sarah left the puddles to dry for 4 hours. Every hour she drew a line around what was left of each puddle as shown in the diagram.



What can Sarah conclude from her observations?

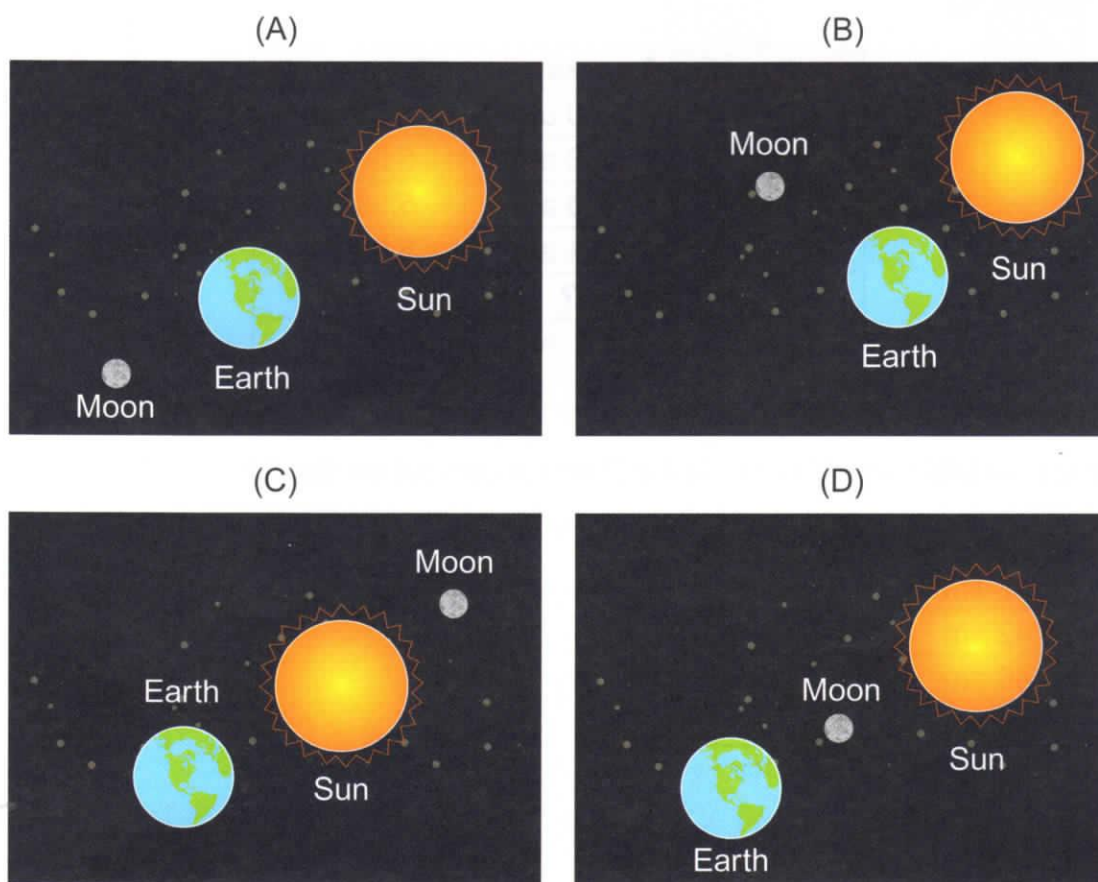
- (A) The liquid in puddle A was water.
(B) Puddle B had more liquid than puddle A.
(C) The liquid in puddle B stopped drying after 4 hours.
(D) The liquid in puddle A dried more slowly than the liquid in puddle B.

9. The diagram compares the features of birds with the features of fish.



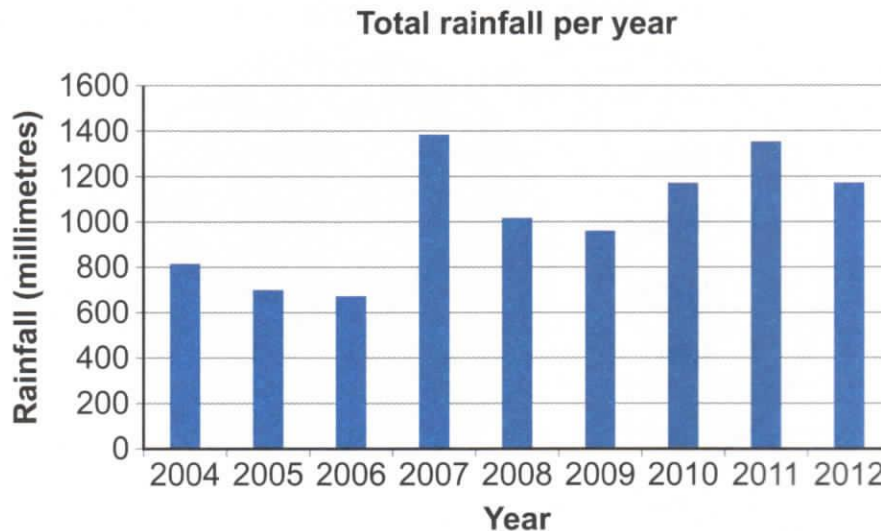
What are two features that birds and fish have in common?

- (A) mouth and tail
(B) fins and teeth
(C) teeth and eyes
(D) feathers and lungs
10. A solar eclipse occurs when the Moon blocks some of the sunlight from reaching Earth.
Which diagram shows a solar eclipse?



(NOT TO SCALE)

11. This graph shows the rainfall per year for a city.



Which years had a total rainfall less than 900 millimetres?

- (A) 2004, 2005 and 2006
(B) 2007, 2008 and 2009
(C) 2004 and 2009 only
(D) 2005 and 2006 only

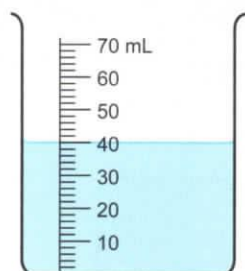
12. The table shows the times that the Sun rose and set during one week. The information for day 5 is missing.

Day	Sun rose at	Sun set at
1	5:50 am	5:56 pm
2	5:48 am	5:57 pm
3	5:46 am	5:58 pm
4	5:44 am	5:59 pm
5	?	?
6	5:40 am	6:01 pm
7	5:38 am	6:02 pm

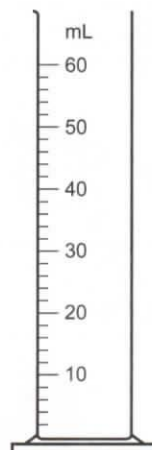
Based on the table, what times did the Sun rise and set on day 5?

	Sun rose at	Sun set at
(A)	5:42 am	6:00 pm
(B)	5:42 am	5:58 pm
(C)	5:46 am	6:00 pm
(D)	5:46 am	5:58 pm

13. Joe had two containers, X and Y.



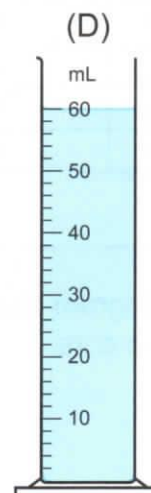
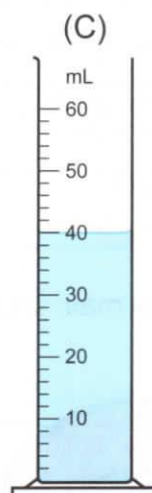
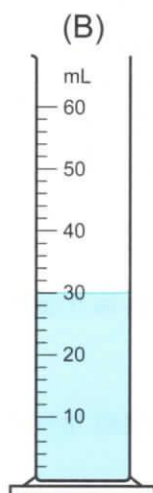
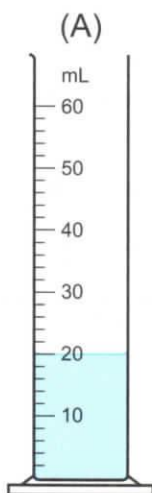
container X



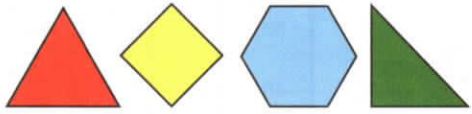
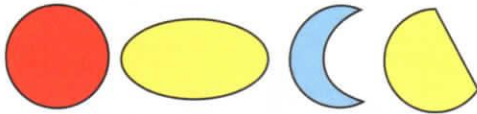
container Y

All the water from container X was poured into container Y.

Which diagram shows the water level in container Y?



14. Mary sorted eight shapes into two groups.

Group 1	Group 2
	

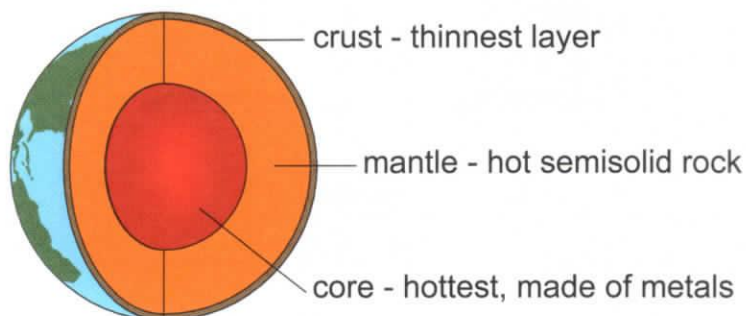
Mary had two more shapes:



Which of these matches each shape with the group using the same rule?

		
(A)	Group 1	Group 2
(B)	Group 1	Group 1
(C)	Group 2	Group 1
(D)	Group 2	Group 2

15. The diagram shows the three main layers that Earth is made up of: the crust, the mantle, and the core.

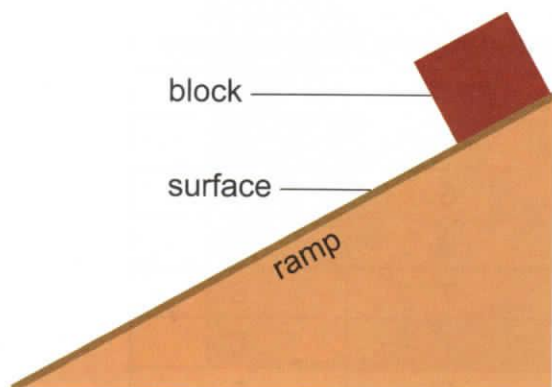


Which of these statements is supported by the information in the diagram?

- (A) The crust is the thinnest and the hottest of the three layers.
- (B) The mantle is hot and in between the crust and the core.
- (C) The core is the hardest and the coldest of the three layers.
- (D) The crust is hot and in between the mantle and the core.

16. Friction is a force that slows down the movement of objects.

Some students recorded the time it took for one block to slide down four ramps. The ramps were the same, except each ramp had a different surface.

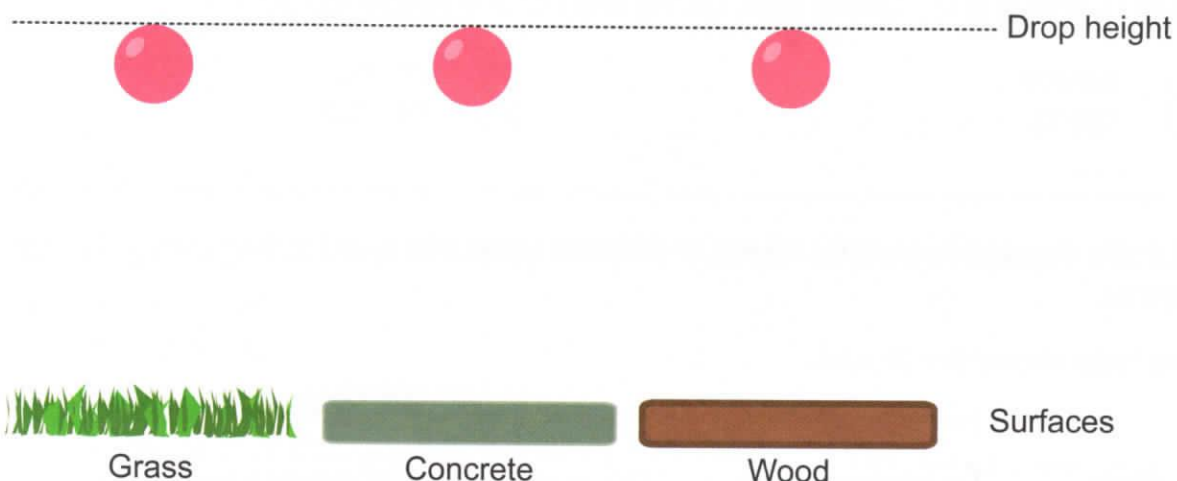


Surface	Time taken for block to slide down the ramp (seconds)
carpet	19
glass	5
wood	10
rubber	12

Which surface caused the most friction with the block?

- (A) carpet (B) glass (C) wood (D) rubber

17. Sam was testing the bounce of a rubber ball. The picture shows three of his tests.



What can Sam find out from these three tests?

- (A) whether the type of ball affects the bounce
(B) whether the type of surface affects the bounce
(C) whether the drop height affects the bounce
(D) whether the ball shape affects the bounce

For questions 18 and 19 use the information below.

The table shows the number of days it takes for different seeds to germinate when planted in soil at different temperatures.

Seeds	Days for seeds to germinate at different temperatures			
	10 °C	15 °C	20 °C	25 °C
onion	13	7	5	4
spinach	12	7	6	5
lettuce	7	4	3	2
cabbage	17	10	7	6
carrot	17	10	7	6
pea	14	9	8	6
tomato	43	14	8	6

18. Which seed will take the longest time to germinate at 15 °C?

- (A) onion (B) cabbage
(C) pea (D) tomato

19. Which of these is the fastest germinating seed at any temperature?

- (A) lettuce (B) tomato
(C) carrot (D) spinach

20. Michelle counted how many weeds of different types she found in four areas of a nature reserve.

The table shows her results.

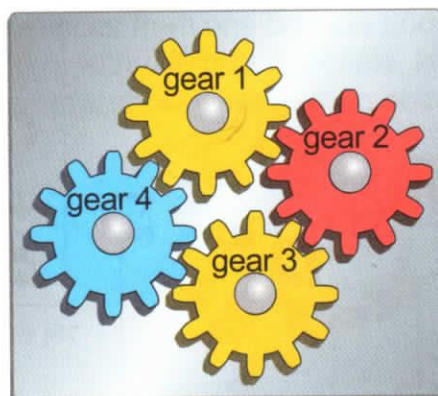
Area	Number of each weed		
	Dandelions	Thistles	Blackberries
1	5	7	0
2	3	7	9
3	6	3	13
4	0	12	0

Which statement about the weeds that Michelle found is correct?

- (A) She found dandelions in all four areas.
(B) She found more blackberries than dandelions.
(C) She found blackberries wherever there were thistles.
(D) She found more thistles than dandelions in each area.

21. Gears are wheels with teeth.

The diagram shows four gears.



Gear 1 is rotating clockwise.

What is the direction of rotation of each of the other gears?

	Gear 2	Gear 3	Gear 4
(A)	clockwise	anticlockwise	clockwise
(B)	anticlockwise	clockwise	anticlockwise
(C)	anticlockwise	anticlockwise	clockwise
(D)	clockwise	clockwise	anticlockwise

22. Scientists made a new material with these features:

- waterproof
- cannot be bent
- does not burn
- can be heated up to 300 °C
- allows heat to pass through it.

Which of these is the most suitable product that could be made using this new material?

- (A) oven gloves
- (B) garden hose
- (C) drinking glass
- (D) saucepan

23. Matt was walking along the sand when he saw these footprints.



Which animal was the last to leave its footprints on the sand?

- (A) cat (B) bird (C) human (D) horse

24. This is a weather map for six towns.



Danny lives in a town that is west of Bogdich and directly north of Dapburg.

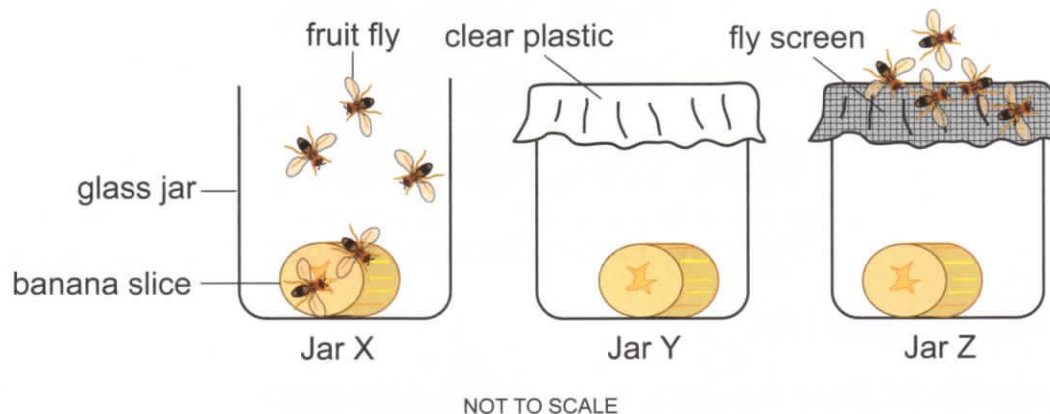
What weather does the map show for Danny's town?

- (A) Sunny (B) Cloudy (C) Rain (D) Lightning

For questions 25 and 26 use the information below.

Nathan wanted to find out how fruit flies find bananas.

Nathan cut three pieces from one banana and placed them into three identical glass jars. He covered two jars with different materials. He placed all three jars together on a table outside. The diagrams show what Nathan observed a short time later.



25. Which condition did Nathan have to keep the same for each jar to make his investigation a fair test?
- (A) the type of cover
(B) the type of fruit fly
(C) the amount of fruit flies
(D) the size of the banana slice
26. Based on these observations, how do fruit flies find bananas?
- (A) by sight
(B) by smell
(C) by either sight or smell
(D) by smell and taste

27. Earthworms usually live in soil that is cool and moist and has air spaces between the soil particles.

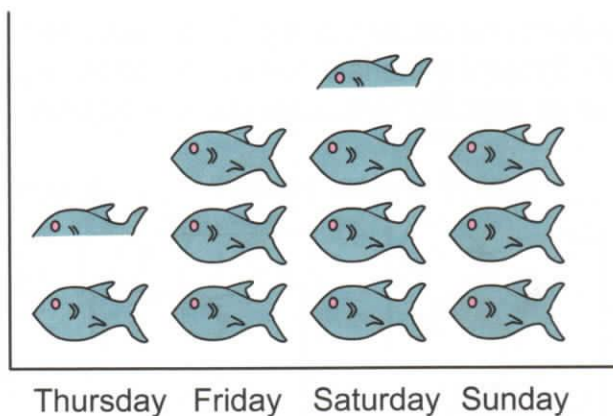
After a day of heavy rain, Josh went outside and looked at his garden. He saw many earthworms on the surface of the wet soil.



What is the most likely reason these earthworms crawled out of the wet soil?

- (A) There was not enough water in the soil.
(B) There were too many earthworms living in the same area.
(C) The earthworms needed air to breathe.
(D) The earthworms needed sunlight to dry their bodies.

28. Sarah counted the number of fish she saw in a river each day over four days. She drew this graph to show the number, but forgot to include a key in her graph.

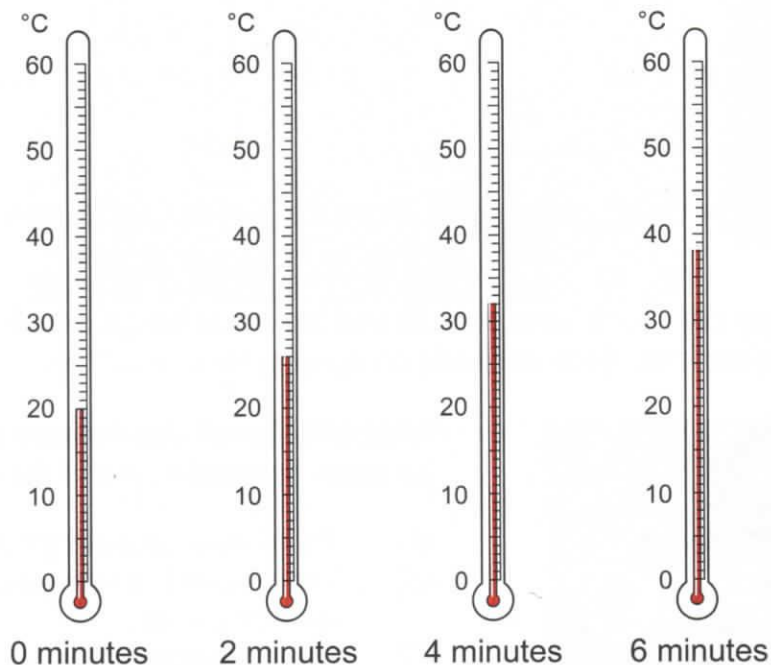


On Thursday she saw 3 fish.

How many fish did Sarah see altogether over these four days?

29. Adam heated some water for eight minutes. He measured the water temperature using a thermometer.

The thermometers show the temperature of the water every two minutes.

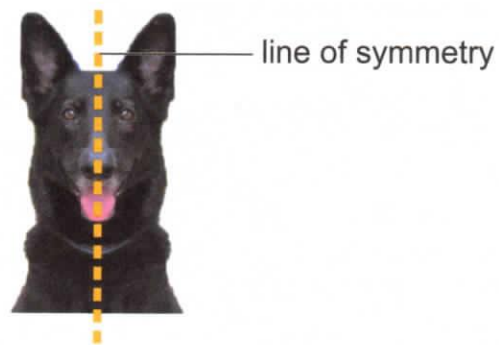


What will the temperature of the water be at 8 minutes?

- (A) 38 °C (B) 42 °C (C) 44 °C (D) 46 °C

30. Many living things are symmetrical.

For example, this picture of a dog has one line of symmetry.



The pictures below show a crab and a flower.



crab



flower

Which option has the correct number of lines of symmetry for each picture?

	crab	flower
(A)	1	2
(B)	2	2
(C)	1	4
(D)	2	4

SOURCES

Question 30

Image of a dog adapted from a photograph by Audrey <https://www.flickr.com/photos/audreyjm529/213989672/> licensed under the Creative Commons Attribution 2.0 Generic license <https://creativecommons.org/licenses/by/2.0>

Yr 3 REACH Science 2019

- | | | |
|-------|-------|-------|
| 1. B | 11. A | 21. B |
| 2. C | 12. A | 22. D |
| 3. D | 13. C | 23. A |
| 4. C | 14. C | 24. A |
| 5. B | 15. B | 25. D |
| 6. D | 16. A | 26. B |
| 7. C | 17. B | 27. C |
| 8. D | 18. D | 28. D |
| 9. A | 19. A | 29. C |
| 10. D | 20. B | 30. C |

THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER

Australia ¹	Year 3
Brunei	Primary 3
Egypt	Year 3
Hong Kong	Primary 3
Indian Subcontinent ²	Class 3
Indonesia	Year 4
Malaysia	Standard 3
Middle East ³	Class 3
New Zealand/ Pacific ⁴	Year 4
Singapore	Primary 2
Southern Africa ⁵	Grade 3



- 1 All international schools registered with UNSW Global (which have an 8-digit school code starting with 46) should sit the papers according to the Australian year levels.
- 2 Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.
- 3 Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.
- 4 Pacific Region: Vanuatu, Papua New Guinea and Fiji.
- 5 Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.



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Please select:

Subject
ScienceYear
2019

Science | 2019 | Year 3

REACH ASSESSMENTS

Questions

The table below shows all questions. The questions answered incorrectly by RYTHAM are shaded. Questions can be sorted by clicking on the table header. Click on the question to view it.

30 questions in total, 30 answered correctly

Development

Summary

Questions

Certificates

↑ Question Number	Area Assessed	Description	Correct Answer	RYTHAM's Answer	AU % Correct
1	Observing/Measuring	Recognise features of animals	B	✓	90
2	Observing/Measuring	Measure the height of an object	C	✓	97
3	Interpreting	Recognise functions of plant parts	D	✓	88
4	Observing/Measuring	Compare the measurements of lengths	C	✓	81
5	Reasoning/Problem solving	Compare the masses of objects by reasoning	B	✓	63
6	Investigating	Recognise the way of collecting data from a group of people	D	✓	84
7	Interpreting	Interpret a two-way table	C	✓	90
8	Predicting/Concluding	Draw a conclusion based on observation	D	✓	72
9	Predicting/Concluding	Identify common features of animal groups from a Venn diagram	A	✓	84
10	Interpreting	Interpret a diagram representing a solar eclipse	D	✓	75
11	Interpreting	Interpret a bar graph	A	✓	74
12	Predicting/Concluding	Make a prediction based on an identified pattern	A	✓	67
13	Observing/Measuring	Identify the equivalent volume in different containers	C	✓	55
14	Predicting/Concluding	Sort shapes based on easily identified characteristics	C	✓	52
15	Interpreting	Interpret a structural diagram	B	✓	73
16	Investigating	Draw a conclusion from the results of an investigation	A	✓	70

17. Investigating [identify the question that can be answered from a set of tests] B ✓ 73

Please select:

Subject

Science

Year

2019



Science | 2019 | Year 3

REACH ASSESSMENTS™

Development

Summary

Questions

Certificates

18

Interpreting

answered from
a set of testsInterpret a
two-way table
and extract
information

D

✓

71

19

Interpreting

Interpret a
two-way table
and extract
information

A

✓

62

20

Predicting/Concluding

Draw a
conclusion
from tabulated
results

B

✓

54

21

Reasoning/Problem
solvingIdentify the
direction of
movement of
rotating gears

B

✓

52

22

Reasoning/Problem
solvingPredict the
potential use
of a product
based on its
properties

D

✓

53

23

Reasoning/Problem
solvingSequence the
relative time of
different
footprints

A

✓

50

24

Interpreting

Read a
weather map

A

✓

53

25

Investigating

Determine
conditions for
a fair test

D

✓

38

26

Investigating

Draw a
conclusion
from
observations
of an
investigation

B

✓

42

27

Reasoning/Problem
solvingDraw an
inference from
information
provided

C

✓

40

28

Predicting/Concluding

Calculate a
total based on
a picture graph

D

✓

41

29

Predicting/Concluding

Extrapolate the
temperature
reading based
on an
identified
pattern

C

✓

37

30

Reasoning/Problem
solvingDetermine the
number of
lines of
symmetry for
given objects

C

✓

40